Cheap Evidence, Hard Errors: Verifying Document Extraction Without Another Generative-Model Call

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Abstract

Intelligent document processing (IDP) pipelines must keep wrong values and broken tables out of downstream systems. Human review and extra language-model checks reduce this risk, but both add cost and latency. We present Evidence-Based Verification (EBV), an accept-or-route layer that judges extracted fields and tables from document-grounded signals and historical field priors, without calling another generative model. EBV auto-accepts well-supported outputs and routes the rest for further review. It compares fields in structured documents with learned page regions, checks fields in unstructured documents with text signals and field priors, and checks tables by their grid structure. We evaluate it on public benchmarks and a proprietary mortgage corpus. Its ability to rank correct above incorrect extractions varies by domain, from near chance on a small NDA set and 0.71 on long charity reports to AUROC 0.96 on SEC filings. On one table benchmark, adding an independent OCR reading to HTML-internal signals raises AUROC from 0.76 to 0.88. On FCC invoices, a threshold set on held-out calibration data auto-accepts 81% of fields with 2.1% test error. The hardest errors are well-formed but wrong, such as a valid value copied from another field or a table whose rows are reordered with no text lost. Cheap evidence is enough for many routine outputs. The remaining cases are where human or generative-model review should be spent.

1 Introduction

Intelligent document processing (IDP) pipelines use large language and vision-language models to extract fields and tables from business documents. Some of these outputs are wrong, so the pipeline must decide which ones to accept automatically and which to send for review. Both kinds of mistake are costly. Accepting a wrong value can corrupt downstream decisions, and sending a correct one for review wastes the reviewer's time. A verifier handles this trade-off by accepting the outputs it can vouch for and routing the rest. The verifier, however, has a cost of its own.

The cheapest signal is the extractor's own confidence, when it exposes one, because it needs no further inference. On document fields, however, token probabilities distinguish correct from incorrect values poorly (Kumar, 2026; Roy et al., 2026). Stronger checks call a generative model again. They re-prompt the extractor (Kadavath et al., 2022), sample it several times (Manakul et al., 2023; Kuhn et al., 2023), or ask a second vision-language model to re-read the page (Jin et al., 2026). Each such call adds cost and latency for every output it checks. Between these two extremes lies the evidence the pipeline has already produced. Prior work uses such evidence for fields or for tables, but rarely for both under one accept-or-route decision.

How many extraction errors can we catch from evidence the pipeline already has, without another generative-model call? Evidence-Based Verification (EBV) starts from the evidence the extraction pipeline has already produced. This includes the page text, word positions, the extracted table's structure and per-field statistics learned from labelled history. For tables, we separately measure the benefit of one additional pass from a lightweight, non-generative OCR reader that serves as an independent witness.

EBV chooses a verifier by document layout and output type, because each carries different evidence. Within one accept-or-route layer, ROI verifies fields in structured (template) documents, where a field sits in a fixed place on the page. The Signal verifier handles fields in unstructured (variable-layout) documents, which have no fixed place, using text signals and field priors. Table-Trust verifies tables by their grid structure and, with the OCR witness,

by whether their text matches the page. EBV detects and routes errors. It does not repair them.

We evaluate EBV on public benchmarks and a proprietary mortgage corpus. The outputs come from openly available extractors, and the errors are the extractors' own, not injected. Cheap evidence is highly predictive on several field benchmarks but near chance on a small NDA set. On tables, an independent OCR reading adds information that HTML-internal signals lack.

Our contributions:

1. **One verification layer across extraction granularities with no additional generative call.** We frame document-extraction verification as one accept-or-route layer over structured fields, unstructured fields and tables. Prior document verifiers either work on fields only or, like RaV-IDP, fall back to a vision-language model (§2).

2. **What each kind of evidence buys.** On SEC filings, a prior on whether each field is usually empty contributes most to ranking. On RD-TableBench, adding an independent OCR witness to HTML-internal signals raises AUROC from 0.76 to 0.88 (§5.2, §6).

3. **Operational limits and deployment lessons.** EBV is deployed in a production IDP pipeline. On proprietary production data covering five document types, it reaches AUROC 0.93–0.99 on fields and catches 31/34 table defects. We report where it over-rejects, where coverage collapses under a high extractor error rate (§5.3), and why well-formed but wrong outputs escape. In the cases we inspected, the document still held evidence that a stronger check could use (§6).

2 Related Work

Confidence for document field extraction. Several systems attach a confidence score to key-information-extraction (KIE) outputs and use it to decide which fields to auto-accept. HYCEDIS (Nguyen et al., 2022) fuses OCR, box-localisation and key–value signals through a conformal predictor tied to one extraction stack. ExtractConf (Kumar, 2026) shows that document- and OCR-grounded signals beat raw token log-probabilities and fuses them with a gradient-boosted model, but its core uses two LLM calls. ConfBench (Roy et al., 2026) benchmarks VLM confidence on degraded RealKIE-FCC documents and reports token-logprob AUROC of 0.59–0.64. Gurram (2026) adds distribution-free guarantees to the per-field decision. Perception–Layout–Validation (Jin et al., 2026) splits confidence into perception, layout and validation channels. It gates straight-through processing with conformal risk control and reports AUROC 0.990–0.996 on FATURA with one VLM extractor, rising to 0.997 when a second VLM is added. Most of these systems handle fields only, and several make an extra LLM or VLM call at decision time. For structured fields, the closest comparison is the layout channel of Perception–Layout–Validation, which scores a value's position against layouts retrieved from similar pages. Our ROI verifier instead uses a fixed region learned for each template field, without layout retrieval or an additional generative-model call. Like ExtractConf, our Signal verifier combines document-grounded features with a gradient-boosted model, but it computes those features without an LLM call.

Table verification. Uncertainty-aware table extraction (Ajayi et al., 2025) combines the uncertainty of the structure-recognition and OCR models to flag likely errors. Ajayi et al. (2024) estimate structure uncertainty by running the model on altered copies of the image. Both rely on the extraction models' own uncertainty rather than checking the extracted content against an independent OCR reading. QUEST (Thomas et al., 2025) learns a model that predicts extraction F1 from structural and contextual features, but uses it to pick pseudo-labels during training, not to route tables at deployment. DEC (Xiao et al., 2026) repairs table-parse errors with an agentic VLM controller, and its gate for accepting a parse is itself a fine-tuned 8B VLM. RaV-IDP (Jha, 2026) is closest in scope, covering text, tables and images. It detects errors by reconstruction and re-extracts with a GPT-4.1 model when fidelity is low. Our evaluation instead isolates what an independent OCR witness adds over HTML-internal signals for the same accept-or-route decision.

Selective prediction and uncertainty. Choosing whether to accept an output or route it for review is the classic selective-prediction problem (Chow, 1970; El-Yaniv and Wiener, 2010; Geifman and El-Yaniv, 2017, 2019). Conformal risk control (Angelopoulos et al., 2024) chooses a threshold that bounds the expected value of a chosen loss,

assuming calibration and test data are exchangeable. In an audit of a deployed on-device model, Pandey et al. (2026) find that cheap signals from a single generation cannot tell its confident errors from correct answers, while repeated sampling can. Reference-free quality estimation in machine translation scores an output from its source alone (Rei et al., 2020, 2022). LLM uncertainty methods sample several answers and check whether they agree, or ask the model to judge itself (Kuhn et al., 2023; Farquhar et al., 2024; Manakul et al., 2023; Kadavath et al., 2022), which costs extra generative passes.

Most of the document-verification methods above cover one output type, rely mainly on the extraction models' own uncertainty or call another large model for a stronger check. EBV instead gives fields and tables a shared accept-or-route decision, using document evidence suited to each output type and, for fields, historical priors, without another generative-model call.

3 Method

3.1 Accept or route

A pipeline reads a document and returns units. A unit is either a field (key–value) or a table (HTML grid). For each unit, EBV outputs accept, meaning it is processed automatically, or route, meaning it goes to a reviewer. A document is accepted only if all its units are, so one doubtful value sends the whole document for review. An upstream label says whether a document is structured, and EBV does no classification of its own. Tables go to Table-Trust, fields in structured documents go to ROI and all other fields go to Signal.

3.2 Structured fields (ROI)

On a fixed template, each field sits in a stable region. Offline, EBV learns a region of interest for each field from the template's labelled training documents. It takes the field's box in each document, removes outliers and aggregates the rest with percentile bounds. It also learns the field's context keywords from neighbouring words, and it grid-searches the weights of a scoring rule with its accept threshold on the same training documents. Online, EBV finds the value's box through the OCR tokens, using the keywords to choose among repeated occurrences. It measures how well the box agrees with the region by overlap, centre distance and intersection over union (IoU), and the scoring rule combines the three. Strong keyword context raises the score by a capped amount, and a blank value is accepted for a field that is usually empty. A non-empty value that cannot be found in the OCR tokens has no spatial support and is treated as least trusted.

3.3 Unstructured fields (Signal)

The Signal verifier combines cheap signals computed from the extracted value, the page text, the other extracted fields and statistics learned from labelled history. None needs a generative model or another OCR pass. Format validity checks that the value has the right form for its field. Value-in-text checks that it appears on the page. Keyword context, source context and label–value distance check that the words around it and its position fit the field. Cross-field copy flags a value repeated from another field, and some domains add a partial-extraction or type-mismatch signal. Two more features describe emptiness. One says whether the value is empty, and the other is the field's historical empty rate, the field prior. For each document type, a random forest and a gradient-boosted classifier are trained offline on these features, and the one with higher accuracy on a held-out part of the labelled training data is used, so test data play no part in the choice. Its probability is the field's trust score. Two deterministic safeguards act on the accept-or-route decision but not on the score. Hard gates route a type mismatch, a blatant cross-field copy or an empty amount. An empty-field rule accepts a blank value when the field prior says the field is usually empty. The implementation also offers a stricter mode that accepts a field only when both classifiers agree.

3.4 Tables (Table-Trust)

Table-Trust asks whether the grid is coherent and whether its text is faithful. Geometry checks logical validity, such as overlapping cells, holes and impossible spans. Structural plausibility checks the filled grid by per-row fill and header-to-column counts. It flags over-segmented or phantom columns and misplaced cells. Text conservation compares the HTML cell text with the OCR witness and flags merged or dropped values. The trust score is a weighted average of the three checks, adjusted by penalties for detected defects, and each failure is reported with a reason code. A table is accepted when its score is at least 85 on a 0–100 scale, the threshold used in production. Without the witness,

the table is scored on geometry and structure alone.

3.5 What runs at decision time

At verification time, EBV makes no generative (LLM or VLM) call. ROI compares the value's position on the page with a stored region, using OCR tokens the pipeline already produced. The Signal verifier runs deterministic checks and one small local classifier. Table-Trust runs structural checks and, for text conservation, the OCR witness, one pass of a lightweight, non-generative OCR engine (docTR) over the table region. The witness is the only optional extra document-reading model, and §5.2 measures what it adds over structure alone.

4 Experimental Setup

Datasets. For structured fields we use FATURA (Limam et al., 2023), synthetic templated invoices, and VRDU Registration-Forms (Wang et al., 2023), real FARA filings. For unstructured fields we use four RealKIE (Townsend et al., 2024) domains, FCC invoices, SEC S-1 filings, UK charity reports and NDAs, which differ in layout and base error rate. For tables we use RD-TableBench (Reducto, 2024). A proprietary corpus of five mortgage document types covers all three verifiers on one pool of documents.

Extractors. We verify the outputs of open, self-hostable extractors, NuExtract3 (NuMind, 2026) for fields and MinerU (Wang et al., 2024) for tables, run locally on the public and proprietary data alike. Their mistakes are the errors EBV must catch.

Correctness labels. To evaluate EBV, each extracted value is labelled correct or incorrect by comparing it with the dataset's gold value. Structured fields are compared by field-aware exact match. Unstructured fields are compared by normalised bidirectional containment, which accepts formatting differences and shortened forms such as "12 Main St" and "12 Main Street". On the unstructured benchmarks we evaluate scalar key–value fields. On the proprietary corpus the gold value is the production system's finalised value, corrected by a person where needed. On FATURA, exact match marks some correctly located values as wrong because only their formatting differs. We count these as correct, and Appendix A gives the strict result. A table is labelled GOOD if its HTML is faithful to the table in the page image and BAD if values are fused, truncated or dropped. On RD-TableBench, whose own ground truth is often wrong, each table was judged against its page crop.

Splits and training. FATURA and VRDU are split 70/30 by document within each template, so they test new documents from known templates, not unseen templates. The RealKIE domains are split by document, with S-1 pages grouped by source filing. The proprietary corpus uses cross-validation grouped by document. ROI and the Signal verifier are trained per document type, so each new type needs labelled data. Table-Trust's weights and threshold were set on production mortgage tables and applied to RD-TableBench unchanged. We evaluate each field and table separately. The document-level rule of §3.1 is a deployment rule and is not evaluated, since the public datasets do not pair fields and tables in the same documents.

Baselines. On the unstructured benchmarks we compare EBV with two signals the pipeline has without an extra model call, measured on the same outputs. These are a per-field emptiness prior and a value-in-source rule, which checks whether the value occurs on the page. For the extractor's token-logprobs we cite published results on comparable fields as context (ConfBench, 0.59–0.64 AUROC).

Metrics. We keep ranking capability apart from the operating decision. Capability is the tie-aware AUROC of the trust score for ordering correct above incorrect outputs. For the Signal verifier this score is the classifier's probability, before the safeguards of §3.3. For unstructured fields we target an error of at most 5% among accepted fields. We draw 20 random 50/50 document-level splits of the evaluation documents into calibration and test halves, set the threshold on each calibration half and apply it unchanged to the test half. We report mean test coverage and mean realised test error, which can exceed the target. For ROI, coverage at 5% error is read from the test risk–coverage curve, so it is a diagnostic rather than a calibrated decision. For tables we report bad-catch, the share of defective tables routed, and false-reject, the share of faithful tables routed, at the production threshold of 85. We report n and base error rate for each benchmark, and document-clustered bootstrap 95% CIs for the main comparisons.

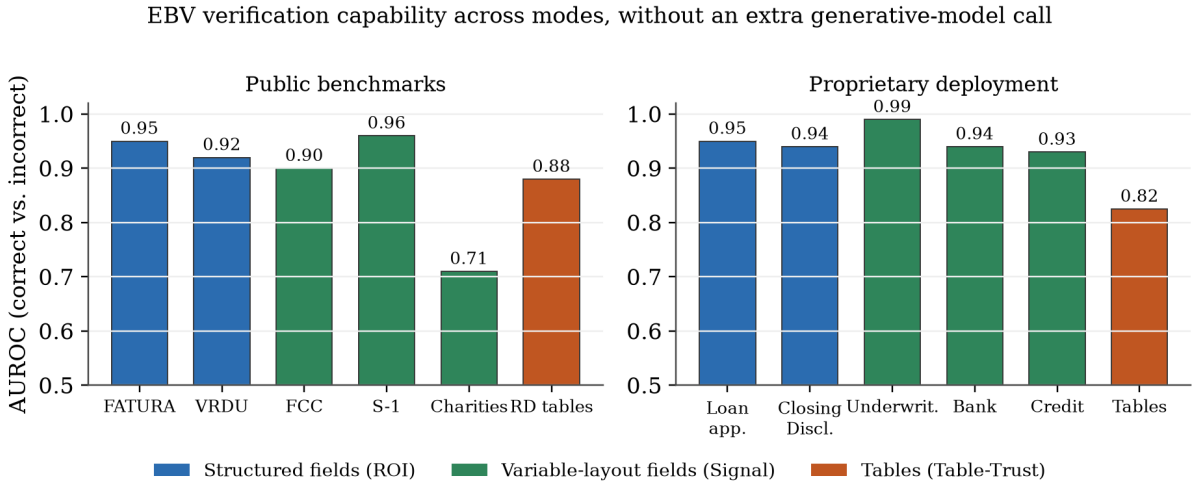


Figure 1: EBV AUROC across the three verifiers, without an extra generative-model call. Left: public benchmarks. Right: proprietary production data. Bars are coloured by verifier (structured fields, unstructured fields, tables).

Benchmark	Verifier	n	base err	EBV AUROC (95% CI)	logprob	prior	presence	coverage (test err)
FATURA	ROI	36,370	23.5%	0.95	—	—	—	70%†
VRDU	ROI	2,647	20.6%	0.92	—	—	—	71%†
FCC	Signal	240	11.3%	0.897 (0.83–0.96)	0.59–0.64‡	n/a	0.53	81% (2.1%)
S-1	Signal	37,277	3.2%	0.960 (0.95–0.97)	—	0.63	0.59	100% (3.3%)
Charities	Signal	1,522	7.0%	0.713 (0.66–0.77)	—	n/a	0.53	79% (5.5%)
NDA	Signal	148	8.8%	0.53	—	—	—	—

Table 1: Field verification. n is the number of scalar field decisions. Bracketed AUROC intervals are document-clustered 95% CIs where reported. ROI AUROC counts every field, and values the verifier cannot locate are ranked lowest. ‡Published ConfBench range on RealKIE-FCC, with different extractors and protocol. For Signal, coverage uses a threshold chosen on calibration halves for a 5% error target, and brackets show the realised test error, which can exceed the target. †ROI coverage is read from the test risk–coverage curve, so it is an optimistic diagnostic, not a deployable operating point. "n/a" marks a prior with no usable variance. On S-1 the base error is already below 5%.

5 Results

5.1 Fields

Wherever a baseline can be measured on the same outputs, EBV ranks correct above incorrect extractions better than the signals the pipeline has without an extra model call (Table 1). The strongest such baseline reaches 0.63, the per-field prior on S-1, and value-in-source reaches 0.53–0.59. On FCC, published token-logprob results lie at 0.59–0.64. Performance varies widely across domains. EBV is near chance on NDAs (0.53), reaches 0.71 on charity reports and ranges from 0.90 to 0.96 on the other public field benchmarks, including 0.95 on FATURA. For context, Jin et al. (2026) report 0.990–0.996 on FATURA with fused perception, layout and validation channels, and 0.997 with an optional second VLM. Their extractors and protocol differ, so this is not a direct comparison. Charity reports are long documents in which the mere presence of a value in the page text says little, and they are the one domain where the realised test error exceeds the 5% target (5.5%). The NDA result is consistent with limited training diversity, since the evaluation holds three field types and only 13 errors among 148 fields.

5.2 Tables: what the OCR witness adds

On RD-TableBench we use the weights and the threshold of 85 set on production mortgage tables, without re-tuning. On the full 460-table set (78% defective), the verifier routes 90% of defective tables. On a 287-table subset (218 defective, 69 faithful), bad-catch is 0.84 at AUROC 0.88. We make no coverage claim on RD, since at this defect rate there is little to auto-accept.

Adding the OCR witness to HTML-internal signals raises table AUROC from 0.76 to 0.88 (Table 2).

Evidence	Additional inference	AUROC
HTML-internal signals	none	0.76
+ OCR witness	1 non-generative OCR pass	0.88

Table 2: What the OCR witness adds (RD-TableBench, 287-table subset). AUROC for ranking faithful above defective tables. The deployed scorer outputs only the combined score, so each row was scored separately. The structure-only row combines four HTML signals (geometry, token shape, structural plausibility and reading order), two of which carry zero weight in the deployed scorer.

5.3 Proprietary production data

On the mortgage corpus, EBV covers all three verifiers (Table 3). On fields the extractor is wrong on 17–43% of values, and EBV ranks them with AUROC 0.93–0.99. Coverage generally falls as the error rate rises, although it also depends on how well the scores separate. On loan applications 86% of fields are auto-accepted, but on bank statements, where the extractor is wrong 43% of the time, only 2% are. Ranking stays strong there, yet little can be auto-accepted, which is the gap between capability and coverage. On tables (n=105, 34 defective), the verifier catches 31/34 defects (0.91; binomial 95% CI 0.76–0.98), including fused late-payment counts, truncated payment histories and dropped rows. The price is a false-reject rate of 24/71 (0.34). Most false rejects are atypical but faithful layouts, mostly on credit reports. Bank statements and atypical tables are where the cheap check should hand off to a stronger reader. Combined table AUROC is 0.82.

6 Analysis: what cheap evidence buys, and the hard errors

An extra reading pays off for table content. The ladder (§5.2) shows *where* evidence matters. Structure catches grid defects but cannot see content substitution, while the independent OCR witness separates a faithful grid from one with merged or dropped text. For one non-generative pass, that is worth +0.12 AUROC over structure alone.

On free-form fields, the empty-field prior carries the most weight. Removing the empty-field prior lowers AUROC from 0.957 to 0.654 (Δ0.30), more than removing any content signal. Many extractor errors are empty outputs on fields that are usually filled (944 of 1,334 Signal errors). This comes from a drop-one ablation on a surrogate tabular classifier that reproduces the deployed Signal probabilities. About 95% of its instances are S-1, so the result describes S-1. The content signals are each informative but overlap heavily. Dropping any single one costs ≈0 AUROC, since the others cover for it. We therefore credit the content signals only for what they add on top of this cheap prior. The claim has limits. On FCC and Charities the prior has no usable variance (Table 1), so the 0.90 and 0.71 there come from content.

The residual errors look right. The verifier catches most of every error family. At its operating point it catches source-absent errors 47→1.00, wrong-value-in-source 313→0.97, empty-output 944→0.79 and cross-field-copy 30→0.83 on Signal, and data-loss 99→0.84, column-explosion 94→0.83 and scramble 22→0.82 on RD tables. What gets through is small, and it has one signature. If we separate genuinely wrong values from cases where the extractor returned nothing, the confidently-wrong residual (outputs that were accepted and scored at least as clean as the median correct output) is 6/397 = 1.5% of non-empty Signal errors (S-1 0%, FCC 0%, Charities 7.0%) and 10.6% [5.8, 17.1] of RD-table errors. In every mode the hardest errors are *plausible-but-wrong*. For fields it is the cross-field copy, a valid value taken from the wrong field and the lowest-caught non-empty family (0.83 on Signal, 0.60 on VRDU). For tables it is the scramble, a reordering that keeps all the content (0.82). The proprietary tables have the same shape. One confident miss drops a document's loan-type checkboxes but leaves a perfectly valid grid, which the structural score trusts completely (trust 100). We adjudicated all 29 confidently-wrong cases (RD 23, Signal 6) and found none source-ambiguous. The information needed to catch each one is in the document. This residual is a cost and coverage boundary that more evidence or a model call could push back.

7 Conclusion

Cheap evidence from the document goes a long way. One verifier covers structured fields, free-form fields and tables without another generative-model call. It ranks correct extractions above wrong ones better than the signals a pipeline already has. One independent OCR reading is the extra cost worth paying, and only for table content. The errors that remain look right. They are values

Doc type	Verifier	n	errors	AUROC	coverage
Loan application	ROI	5,970	1,018 (17%)	0.95	86%
Closing disclosure	ROI	3,300	574 (17%)	0.94	81%
Underwriting findings	Signal	5,970	1,602 (27%)	0.99	75%
Bank statements	Signal	1,330	567 (43%)	0.94	2%
Credit reports	Signal	3,204	1,133 (35%)	0.93	34%
Tables	Table-Trust	105	34 defective	0.82	bad-catch 0.91; FRR 0.34

Table 3: Proprietary production data. Field coverage is at a 5% error target. Loan application and underwriting findings share one document pool, hence the same n. FRR is the false-reject rate.

copied from the wrong field and tables reordered without loss of text. The evidence to catch them is in the document. Catching them needs more evidence or a model call.

Limitations

(i) On the proprietary corpus, field ground truth is the production pipeline's human-verified value. Where production routing and our verifier respond to related signals, this may flatter agreement on the accept side. An independent audit of fields accepted as correct is future work. (ii) We do not compare against an equal-cost LLM judge, since our claim is about catching errors *without* such a call. We also do not re-run the extractor's token-logprobs on our splits and cite published values instead (§4). A matched logprob column would be a natural addition. (iii) The cross-domain results evaluate the verifier on new datasets and need labelled data for each type, so they are not a frozen-model test of distribution shift. (iv) On free-form fields, much of the deployed signal is an empty-field prior (§6). We report this openly and do not present it as content understanding. The mix of error types is a property of this extractor. A stronger extractor would shift mass toward plausible-but-wrong errors. (v) The detectability account is empirical and relative to the evidence we allow. It describes which errors that evidence separates, not a fundamental limit. (vi) Faithful (defect-free) tables are scarce in public benchmarks, so the table conclusions rest on modest counts of faithful tables. Section 5 gives the confidence intervals. (vii) We report per-unit decisions. The document-level conjunction (§3.1) can be configured per deployment by field criticality, and we do not report a single document-level auto-approval rate.

Ethical Considerations

EBV reduces the burden of human review by auto-accepting outputs it can support, and it never takes the human out of the loop for those it cannot. The document-level conjunction is deliberately conservative. The proprietary corpus contains personal and financial information and was handled under its owner's data agreements. We release only aggregate metrics, with no document content, field values or decision logs. Because routing shapes downstream workflows, EBV is an aid for selective acceptance and not an autonomous decision-maker. Its false-accept and false-reject behaviour (§5–§6) must be re-validated for each deployment, vendor and document type before any automation threshold is set.

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A FATURA correctness scoring

On FATURA, strict exact scoring marks as wrong some spatially correct values whose string differs (type-A, 20.8% of raw errors). We relabel these as correct by containment and keep the genuine in-place errors (type-C, 3.8% of predictions). This puts true capability somewhere in [0.865, 0.994], with strict exact scoring at the lower end and containment at the upper end. The FATURA base error in Table 1 (23.5%) uses this relabelling. Raw strict-exact error is 29.7%.